

EDUCATION

- **University of Wisconsin, Madison** September 2024 - Present
Bachelor of Engineering - Computer Engineering

SKILLS SUMMARY

- **Programming Languages:** Python, Java, JavaScript, Kotlin, C, C++, HTML, Linux, SQL, NoSQL, Assembly, Verilog, Perl
- **Frameworks/Libraries:** React, NodeJS, Embedded C, Flask, OpenCV, pyTorch, Ollama, Kafka, LabVIEW, Codex, Claude Code
- **Technologies:** TestStand, MongoDB, AWS, SolidWorks, Android Studio, Cadence Capture, DraftSight, Altium Designer, MCP
- **Areas of Expertise:** Data Structures & Algorithms, Computer Architecture, Circuits & Electronics, Microprocessor Systems, Chip Design, HDL, FPGA/ASIC Architecture, System Verilog, RTL, PSoC6, Cortex & ARM Based Systems

WORK EXPERIENCE

- **Plexus Corporation** Neenah, WI
Functional Test Engineering Design Intern May 2026 - September 2026
 - Delivered **2 critical FT Systems** for major **aerospace and defense clients**; worked directly with clients on specification design for **\$100K+ systems** supporting **\$200M+** in business with Plexus.
 - Served as **lead designer** for a **pharmaceutical organization** using **Cadence Capture**, creating component symbols and system/custom **PCBA schematics** from design specifications.
 - Utilized **DraftSight** for accurate, efficient **CAD design of high-level assemblies and PCBAs**; supported **specification, fabrication, integration, and shipment** across the system development lifecycle.
 - Increased **software design phase** productivity by **3x** while developing **TestStand** sequences and **LabVIEW drivers** for automated **Keysight/Fluke instrumentation** and product interfaces in **40U FT Systems**.
 - Resolved **IC and board-level sensitivity issues** through **root cause analysis**, instrument measurements, and **circuit analysis** during system integration; completed **technical documentation** and prepared systems for factory-floor deployment.
- **Neuron Edge AI (SMB Service) (Automotive Suite)** San Francisco, CA
AI/ML Research - Developer May 2025 - September 2025
 - Built an SMB revenue platform with **Flask, MongoDB, and REST APIs** to automate revenue tracking and reporting, streamline transaction processing, and improve visibility into financial performance.
 - Integrated **ML models** and used **Ollama** to run **Llama 3.3** and **DeepSeek R1** for revenue forecasting and transaction anomaly detection, turning financial data into actionable insights for SMB clients.
 - Developed automotive **microservices** for telemetry, diagnostics, and predictive analytics to support **vehicle performance monitoring and fault analysis**; used **containerization, CI/CD, and integration testing** for reliable deployments.
- **Baja SAE Team (Team Baja)** Madison, WI
Team Captain - Sensory Input August 2024 - Present
 - Led the Sensory and Data subteam in designing and implementing a real-time vehicle telemetry system for off-road navigation using **Arduino, Raspberry Pi**, and multiple sensors.
 - Developed **Arduino-based telemetry** using **gyroscope and passive infrared sensors** on tires to track traction, braking, and suspension dynamics in real time for data-driven **off-road performance optimization**.
 - Managed task **delegation** across multidisciplinary teams, overseeing sensor integration, **CAD modeling, 3D printing, and CNC machining for sensor arrays**, ensuring seamless collaboration and execution.
 - Developed an interactive **React-based GUI dashboard** to monitor tire wear patterns and traction loss in real time, improving **predictive maintenance** and vehicle performance analysis.
- **ACM Sacramento Chapter (FCUSD Panel)** Sacramento, CA
Lead Project Designer/Engineer May 2024 - September 2024
 - Designed a custom **PCB in Altium Designer** for **BART** passenger monitoring, integrating an **ultrasonic and infrared sensor array** with **Raspberry Pi** and **Python** for real-time passenger detection and traffic tracking.
 - Implemented **AWS CloudWatch** logging to timestamp sensor readings at station stops and regular intervals, capturing passenger traffic patterns over time to support **historical analysis and predictive modeling**.
 - Built an interactive **Python/Matplotlib GUI** with **heatmap analysis** to visualize passenger flow, peak demand, and train occupancy, translating sensor data into **predictive insights** for train scheduling.
 - Trained **ML models on 3 months of data** to inform scheduling for **204,800 weekly riders**, helping increase **passenger throughput by 14%** and reduce **waiting times by 8%**.
- **Data Nimbus** San Francisco, CA
Student Software Engineer March 2023 - May 2024
 - Designed and implemented **RESTful APIs** with **Node.js** and **Express** to handle over **10,000 daily financial transaction lookups**, reducing average response times from **500ms to 300ms (40% faster)**.
 - Integrated **OAuth 2.0** and **JWT-based** authentication/authorization to secure sensitive financial data, preventing unauthorized access and ensuring **PCI-compliant** data handling.
 - Optimized **MongoDB** queries by using **compound indexes** and **aggregation pipelines**, cutting query latency in half, enabling the platform (testing environment) to scale to **2x** user load without degradation.
 - Built and executed **Postman test suites** and automated **CI/CD testing pipelines**, reducing production defects by **50%** and ensuring consistent reliability across deployment cycles.

AWARDS & LANGUAGES

- **Awards:** Robotics Judge's choice award winner, Innovate award by Raytheon, Honor Roll (Academics)
- **Languages:** English, Spanish, Hindi, Telugu